



Fresh Supplies

Predictive Analytics to Cut Post-Harvest Losses

THE PROBLEM

A quarter of every harvest never reaches a buyer.



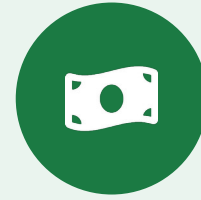
25%

of harvested produce spoils before sale



Nearest,

not best-paying market, is the default
choice



KES 5M+

in lost value every year, before this
problem

WHY THIS MATTERS NOW

Every season this goes unsolved, it compounds.

Spoilage isn't a one-time loss. It's a recurring tax on every harvest, every season, compounded by produce routed to the nearest market instead of the one paying the most.

The cost, in one number

KES 5,000,000+

lost to spoilage alone, every single year, on current harvest volumes



One extra piece of information, at the moment it matters.



1. Log the harvest

Crop, quantity, and basic storage conditions — entered on a phone app or a simple text menu.

2. Predict the risk

The system estimates spoilage risk for that specific batch, in seconds.

3. Recommend the market

One sentence: which nearby market pays the most before it spoils.

No change to how produce is grown, stored, or transported — only better information.

Works on any phone — smartphone or not.



Mobile App

For farmers and drivers with a smartphone — offline-first, so it works even without a live data connection, syncing automatically once back in range.



Text-Message Menu

For any phone, on any network, with no app install and no data plan required — the same predictions and market recommendation, one text at a time.

Two signals, combined into one answer.



Storage & handling risk

How the batch has been stored and handled since harvest — temperature, time elapsed, and how fragile that crop typically is.



Transit risk

How the trip to a given market — distance, road conditions, expected duration — is likely to affect that same batch.



Combined into: what's it worth, where?

The two risks combine into one spoilage estimate per market, ranked by the money actually retained after loss — not just the closest option.

The math, in full.

	Sold Volume	Avg Price/kg	Revenue
Baseline (today)	375,000 kg	KES 40.00	KES 15,000,000
With Fresh Supplies	412,500 kg	KES 43.20	KES 17,820,000

Annual Financial Benefit = **KES 2,820,000 / year**

526%

Year-1 ROI

~1.9 mo

Payback Period

KES 2.52M

Year-2+ Annual Savings

Assumptions

- 500,000 kg annual harvest
- KES 40/kg farm-gate value
- 25% current spoilage rate
- 30% relative spoilage reduction
- 8% price lift from better market match

Conservative, model-based estimates — see Risks (slide 9) for validation plan.

CURRENT STATUS

This isn't a concept. Most of it is already running.



Backend platform and prediction engine — live



Web dashboard for cooperative and logistics staff — live



Farmer mobile app — in active development



Text-message channel — designed, build starting next

Honest caveat: the prediction model is currently calibrated on realistic, but synthetic, data. It has not yet been field-validated on this cooperative's own harvests — which is exactly what the pilot on slide 11 is for.

The two risks worth the board's attention.



Unvalidated real-world accuracy

Projections come from a calibrated model, not yet tested on this cooperative's actual harvests — year-one results could differ from forecast.

Mitigation: Run a 90-day pilot at one collection point first. Measure actual outcomes against the forecast before any cooperative-wide commitment.



Farmer adoption

Some farmers have limited smartphone access or connectivity, which could limit reach if the app were the only channel.

Mitigation: The text-message channel (slide 5) requires no smartphone, no app, and no data plan — every farmer can participate on any phone.

Three phases, each gated by proof, not hope.



MONTHS 1-3

Phase 1 — Pilot

One collection point. Real farmers, real harvests. Measure actual vs. predicted outcomes.



MONTH 4

Phase 2 — Evaluate

Compare pilot results to this forecast. Re-price the business case with real numbers before scaling.



MONTHS 5+

Phase 3 — Scale

Cooperative-wide rollout, plus the text-message channel reaching farmers without smartphones.



The Ask

Approve a 90-day pilot at one collection point.

Full first-year investment: KES 450,000. Projected payback: under 2 months, even at pilot scale. Full cooperative-wide commitment comes only after the pilot proves the numbers on real harvests.

GO — phased. Low cost. Fast proof. Full rollout only after validation.



Thank you.

Questions and discussion.